

fixed electron beam gun item table

ID	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
FG-1	Cathode	Single-material emitter when isolated; cathode assembly is multi-part	Emit thermionic electrons as the electron-beam source.	Emitter foil, filament, or crystal; clamp or holder interface; heater contact.	Tantalum foil; tungsten filament; LaB ₆ cathode for long-life option.	This is the BINP EPAC08 Figure 3 legend and cathode description. Page p.2. Quote: "5-cathode."	The BINP paper directly specifies the cathode material and thickness; the NASA AIAA space EBF ³ paper supports LaB ₆ for longer operating life. Page p.2 quote: "tantalum foil 0.1÷0.2 mm." Page p.3 quote: "lanthanum hexaboride (LaB6) cathode."
FG-2	Anode	Single-material electrode or assembly if using insert/cooling	Provide accelerating field and beam exit aperture after the cathode/control-electrode region.	Anode body, aperture, mounting face, optional replaceable aperture insert.	Molybdenum, tungsten, tantalum, titanium, stainless steel; OFHC copper heat spreader if needed.	This is the BINP EPAC08 Figure 3 legend and electron-gun description. Page p.1–2. Quote: "7-anode."	This is Daiwa Techno's electron-microscope aperture material page; it supports refractory and stainless materials for precise beam apertures. Web page. Quote: "Molybdenum, tantalum, tungsten."
FG-3	Control electrode / Grid / Wehnelt	Single-material precision electrode	Control beam current, cut-off behavior, and initial beam shape near the cathode.	Shaped electrode body, aperture, mounting interface, bias connection.	Molybdenum, tantalum, Kovar, 316L stainless steel; high-temperature versions may use refractory metals.	This is the BINP paper's Figure 3 legend and electron-gun description. Page p.1–2. Quote: "control electrode."	This is Daiwa Techno's electron-gun parts page, listing materials used for electron-gun components. Web page. Quote: "Material: Kovar, Mo, Ta."
FG-4	Control-electrode insulator	Single-material part or small assembly	Support and electrically isolate the control electrode so it can hold a bias relative to the cathode/anode.	Ceramic standoff, insulating sleeve, mounting washer, optional metalized end.	Alumina ceramic, steatite ceramic, glazed alumina; optional Kovar or stainless-steel interface.	This is the BINP EPAC08 Figure 3 legend. Page p.2. Quote: "3-insulator control electrode."	This is CeramTec's feedthrough material page, used here as ceramic-insulator material support. Web page. Quote: "alumina ceramics."
FG-5	Screen electrode / Boundary electrode	Single-material precision electrode	Shape the electric field near the cathode and reduce unwanted side or edge emission effects.	Shaped electrode, aperture edge, mounting interface, electrical connection.	Molybdenum, tantalum, tungsten, Kovar, 316L stainless steel.	This is the BINP EPAC08 Figure 3 legend and gun description. Page p.1–2.	This is Daiwa Techno's electron-gun parts page, used as electron-optics material

ID	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
						Quote: “screen electrode.”	support. Web page. Quote: “Molybdenum (Mo).”
FG-6	Dynamic magnetic lens	Assembly	Dynamically adjust focusing conditions so the beam can remain focused over changing work distance or process needs.	Copper coil, pole pieces, magnetic yoke, bobbin, insulation, mounting structure.	Copper or OFHC copper coil; soft iron pole pieces/yoke; alumina or polyimide insulation.	This is the BINP EPAC08 paper describing static and dynamic electromagnetic focusing. Page p.1–2. Quote: “dynamic electromagnetic lens.”	This is an Iowa State electron-lens teaching PDF describing magnetic-lens construction. PDF. Quote: “soft-iron pole-pieces” and “copper coils.”
FG-7	Main magnetic lens	Assembly	Provide primary focusing of the electron beam to the desired spot size or working distance.	Copper coil, pole pieces, magnetic yoke/shell, bobbin, insulation, mounting interface.	Copper or OFHC copper coil; soft iron or Fe-Co pole pieces; ceramic or polyimide insulation.	This is the BINP EPAC08 Figure 3 legend and optical-system description. Page p.2. Quote: “10-main magnetic lens.”	This is a MyScope/TEM educational source describing electromagnetic lenses. Web page. Quote: “a coil of copper wire.”
FG-8	Two-coordinate deflection coils	Assembly	Deflect the beam in two coordinates for scanning, beam placement, or beam motion on the workpiece.	X/Y coil pairs, coil former, insulation, leads, mount, optional magnetic yoke.	Copper or OFHC copper wire; polyimide or ceramic insulation; stainless or ceramic support.	This is the BINP EPAC08 Figure 3 legend and beam-scanning description. Page p.1–2. Quote: “two coordinates deflected coils.”	This is the Iowa State electron-lens PDF used as general magnetic-coil material support. PDF. Quote: “copper coils.”
FG-9	Pick-up of beam boundary	Assembly	Detect when the beam reaches or exceeds the allowed boundary, supporting beam-deviation protection and interlock.	Collector electrode, insulating mount, shielded signal lead, feedthrough/interface.	Graphite, OFHC copper, molybdenum, or tungsten collector; alumina insulator; stainless support.	This is the BINP EPAC08 Figure 3 legend and exit-protection description. Page p.1–2. Quote: “pick-up of beam boundary.”	This is a JACOW/PAC Faraday-cup design paper, used as material support for beam-intercepting diagnostics. PDF. Quote: “graphite and copper pieces.”
FG-10	Pick-up of secondary electrons	Assembly	Detect secondary or reflected electrons from the workpiece or nearby structures for process monitoring.	Collector plate or ring, bias electrode if needed, insulator, shielded signal lead, feedthrough.	316L stainless steel, molybdenum, or OFHC copper collector; alumina insulator.	This is the BINP EPAC08 Figure 3 legend and reflected-electron monitoring description. Page p.1–2. Quote: “pick-up of secondary electrons.”	This is the JACOW/PAC Faraday-cup paper, used as collector-material support for charged-particle diagnostics. PDF. Quote: “beam dump (stainless steel pieces).”
FG-11	Trajectory corrector	Assembly	Correct beam-axis alignment after the anode so the beam passes through the optical axis and lens center.	Magnetic corrector coils or electrostatic correction plates, insulation, mount, signal/power leads.	Magnetic option: copper coils, soft iron yoke. Electrostatic option: molybdenum or stainless-steel electrodes,	This is the BINP EPAC08 paper describing the beam-axis correction system after the anode. Page p.1–2. Quote:	This is the Iowa State electron-lens PDF, used as material support if the corrector is magnetic. PDF.

ID	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
					alumina insulators.	"15-trajectory corrector."	Quote: "copper coils."
FG-12	High-voltage input	Assembly	Bring accelerating high voltage from the HV tank/cable into the gun-side high-voltage region.	Central conductor, ceramic insulator, metal flange/housing, shield, cable-side termination.	Alumina ceramic-to-metal feedthrough; Kovar or stainless housing; copper or molybdenum conductor.	This is the BINP EPAC08 Figure 3 legend. Page p.2. Quote: "14-high voltage input."	This is CeramTec's ceramic-to-metal feedthrough material page. Web page. Quote: "Metals include Kovar®, stainless steel."
FG-13	High-voltage insulator	Single-material part or ceramic-metal assembly	Electrically isolate the high-voltage gun region from grounded structures.	Ceramic insulator body; optional metallized ends or metal collars.	Alumina ceramic; glazed alumina; optional Kovar or stainless-steel metal interfaces.	This is the BINP EPAC08 paper's Figure 3 legend for the electron-optical system. Page p.2. Quote: "2-high voltage insulator."	This is CeramTec's ceramic-to-metal feedthrough material page; it describes alumina insulators and compatible metals. Web page. Quote: "high-purity, high-strength, low-loss alumina ceramics."
FG-14	Cathode heater leads	Assembly	Deliver heater current to the cathode so thermionic emission can be maintained.	Heater leads, hot-side conductor, current bus, ceramic beads/sleeves, mounting hardware.	Tungsten, rhenium-tungsten, tantalum, or molybdenum hot leads; OFHC copper bus; alumina insulation.	This is the BINP EPAC08 paper describing cathode heater requirements in the high-voltage unit. Page p.3. Quote: "Cathode heater with stabilized current."	This is JEOL's thermionic-emission gun glossary, supporting tungsten and LaB ₆ cathode/heater-related thermionic sources. Web page. Quote: "A tungsten (W) filament."
FG-15	Cathode cartridge / Emitter holder	Assembly	Hold, locate, and allow replacement of the cathode without re-aligning the beam axis.	Emitter holder, clamp, locating seat, electrical contact, ceramic locator, fastening hardware.	Molybdenum, tantalum, tungsten, Kovar, graphite/carbon holder, alumina locator.	This is the BINP EPAC08 paper; it explicitly describes a pre-adjusted cathode cartridge. Page p.2. Quote: "preliminary adjusted cartridge."	This is Daiwa Techno's electron-gun parts page, used as material support for precision electron-gun parts. Web page. Quote: "Kovar, Mo, Ta."
FG-16	Cathode radiation shield	Assembly or multi-layer sheet set	Reduce radiative heat loss from the cathode and protect nearby electrodes/insulators.	One or more thin refractory-metal shields, spacers, small support clips.	Molybdenum foil, tantalum foil, tungsten foil, polished refractory sheet.	This is inferred from the BINP cathode/heater arrangement and high-temperature thermionic operation; Figure 3 does not list this as an independent item. Page p.2. Quote: "direct heated cathode."	This is BARC's electron-gun chapter, describing common high-temperature emitter materials. Page p.8. Quote: "Tantalum and tungsten are widely used."

ID	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
FG-17	Gun column	Assembly	Mechanically support electrodes, lenses, diagnostics, HV input, and the gun-to-chamber interface.	Gun body, column tube, flanges, mounting datum, internal brackets, grounding structure.	316L stainless steel, 304L stainless steel, aluminum alloy, titanium alloy; local OFHC copper heat spreader if needed.	This is NASA's AIAA paper on a flight-weight EBF ³ system; it describes the fixed gun inserted through the chamber top. Page p.3. Quote: "inserted through the top."	This is a USPAS accelerator-vacuum materials lecture supporting stainless steel as a vacuum construction material. PDF. Quote: "Stainless steels are most commonly used."
FG-18	Gun-side oil tank	Assembly	Provide gun-side high-voltage insulation volume near the cathode/HV region.	Oil tank shell, lid, oil volume, seals, internal support, grounding interface.	Stainless steel or aluminum alloy shell; silicone oil; oil-compatible seals.	This is a BINP EPAC08 conference paper describing a 60 keV / 30 kW electron beam facility and its Figure 3 electron-optical system. Page p.2. Quote: "1-oil tank."	The same BINP facility paper states the insulating fluid used in its high-voltage oil tank. Page p.3. Quote: "silicon oil."
FG-19	Gun-side sensing and signal wiring	Assembly	Bring gun-side temperature, current, beam-diagnostic, and interlock signals to controls.	Sensor pickup, shielded wires, ceramic feedthrough, connector, grounding/shield termination.	Ceramic-insulated thermocouple, RTD, shielded copper wire, Kapton/polyimide wire, alumina feedthrough.	This is the BINP EPAC08 paper describing monitored parameters and gun-related currents. Page p.3. Quote: "beam current, cathode heat current."	This is CeramTec's high-alumina ceramic terminal page, supporting electrical/signal passage through a case or housing. Web page. Quote: "pass electrical power or signal."